

International AS • Edexcel • Chemistry

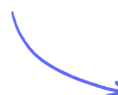
 5 mins 4 questions

Multiple Choice Questions

Atomic Structure

Sub-Atomic Particles / Isotopes & Mass Spectra

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Total Marks

/5

- 1 The numbers of subatomic particles present in four species **W**, **X**, **Y** and **Z** are given in the table.

Species	Number of protons	Number of neutrons	Number of electrons
W	19	20	18
X	19	20	19
Y	20	20	18
Z	20	22	20

Which of these species are isotopic?

- A. W and X**
- B. W and Y**
- C. X and Z**
- D. Y and Z**

(1 mark)

- 2 Iodine exists as one isotope with mass number 127.

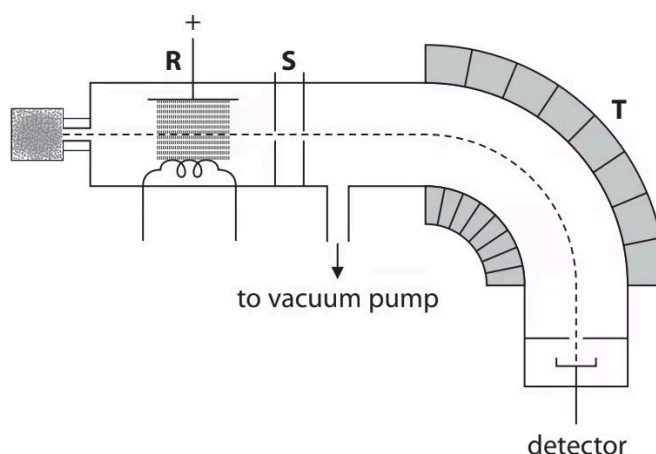
Chlorine exists as two isotopes with mass numbers 35 and 37.

How many molecular ion (ICl_3^+) peaks are there in the mass spectrum of ICl_3 ?

- A. 2**
- B. 3**
- C. 4**
- D. 5**

(1 mark)

3 (a) The diagram shows a mass spectrometer.



Which process occurs in region **R**?

- A.** the sample is vaporised using a heater
- B.** electrons are removed from molecules or atoms and positive ions are formed
- C.** electrons are added to the molecules or atoms and negative ions are formed
- D.** ions are accelerated by an electric field

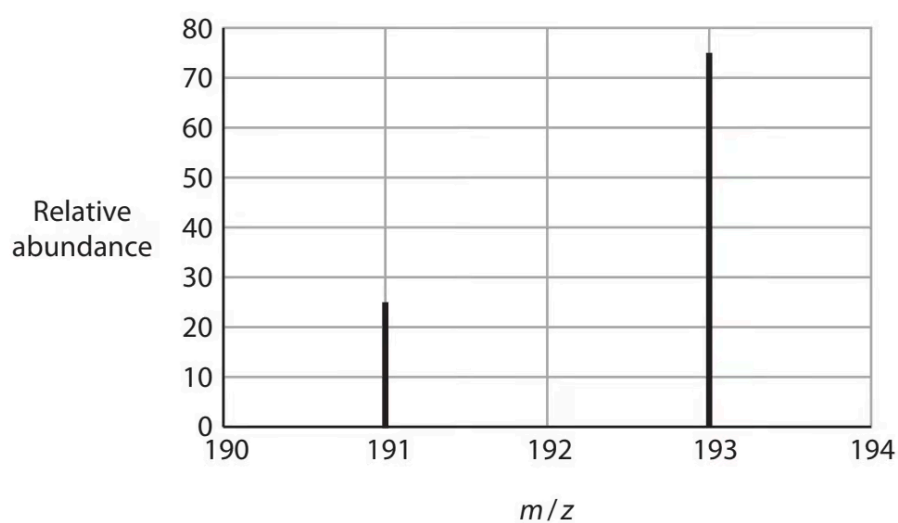
(1 mark)

(b) Which statement is correct for region **T**?

- A.** ions with a greater mass have a smaller deflection
- B.** ions with a greater mass have a greater deflection
- C.** ions with a greater charge have a smaller deflection
- D.** ions are speeded up by a magnetic field

(1 mark)

4 The mass spectrum of a sample of an element has only two peaks.



What is the approximate relative atomic mass of the element in this sample?

- A. 191.5
- B. 192.0
- C. 192.5
- D. 193.0

(1 mark)